

Unit 13, Lesson 3: Angle Relationships in Triangles

Benchmark

MA.8.GR.1.5 Solve problems involving the relationships of interior and exterior angles of a triangle.

Learning Targets

- ☐ I can identify the relationships of interior and exterior angles of a triangle.
- ☐ I can write and solve algebraic equations to solve mathematical problems involving angle relationships of triangles.

13.3.1 Warm-Up: Math Talks

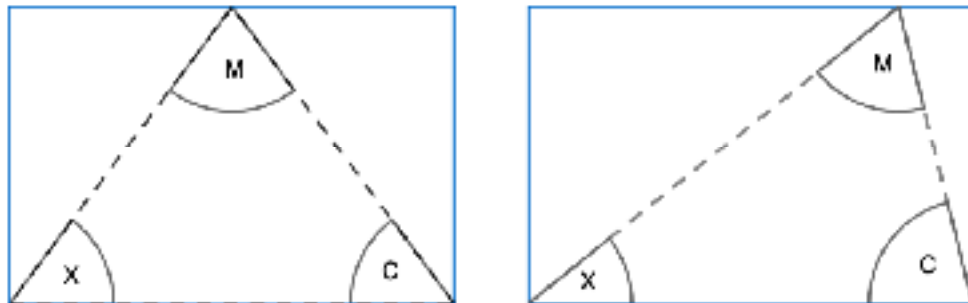
1. Angle measures are often used in construction projects. Special saws, like the miter saw shown, are used to make precise cuts into wood and other raw materials.



- a. What is something you notice about the miter saw?
- b. What is something you wonder about the miter saw?

13.3.2 Exploration: Interior Angles of a Triangle

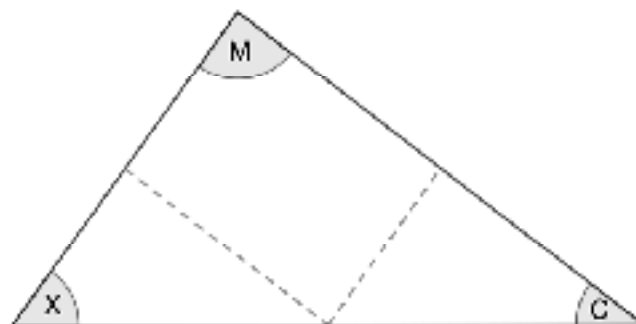
1. Complete the following using a piece of rectangular paper.
 - a. Use a straightedge to draw diagonals from the bottom two corners to intersect on the top side of the paper. Two possible examples are shown.



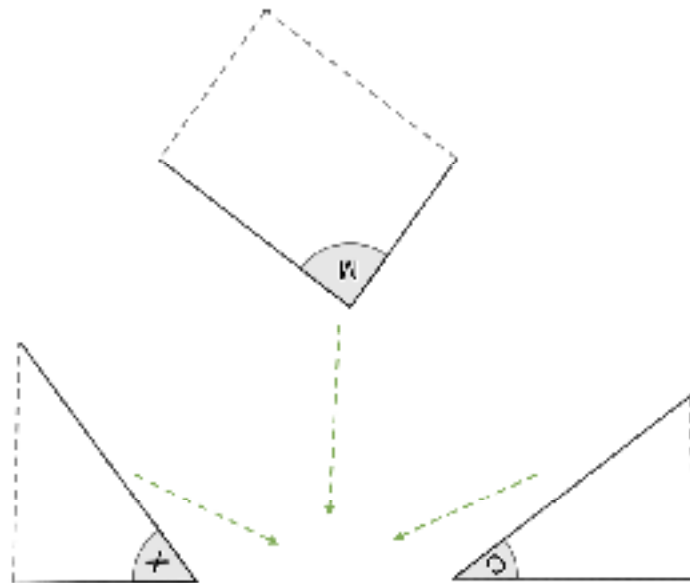
- b. Label the angle formed where the lines you drew intersect *M*. Then, label the other angles in the triangle *X* and *C*, as shown.

These angles are called the **interior angles** of the triangle because they are *inside* the triangle.

- c. Cut out the triangle.
 - d. Cut or tear the triangle so that each angle is separated. An example of where to cut is shown.



- e. Rotate and slide the labeled angles to form a set of three adjacent angles. The image shows the beginning of this process with arrows indicating where the angles should continue to slide.



- f. What do you notice?

- g. Compare your findings with your partner.

2. Complete the conjecture about the relationship of the three interior angles of a triangle.

Adjacent angles that form a straight line have a sum of

- ☐ 90°.
- ☐ 180°.
- ☐ 360°.

Based on the findings from the Exploration, the three interior angles of a triangle

sum to

- ☐ 90°.
- ☐ 180°.
- ☐ 360°.

13.3.3 Guided Practice: Interior Angles of Triangles

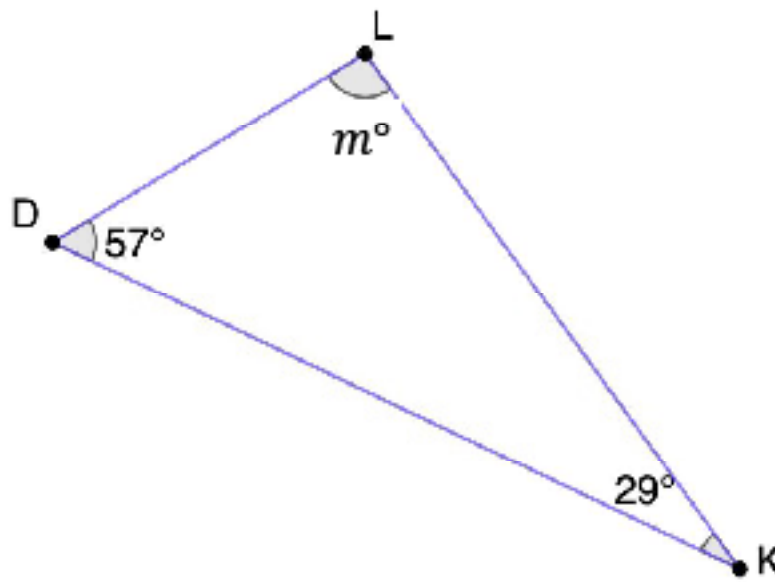
The relationship of the interior angles of a triangle can be used to solve for unknown angle measures in a triangle.



Triangle Sum Theorem

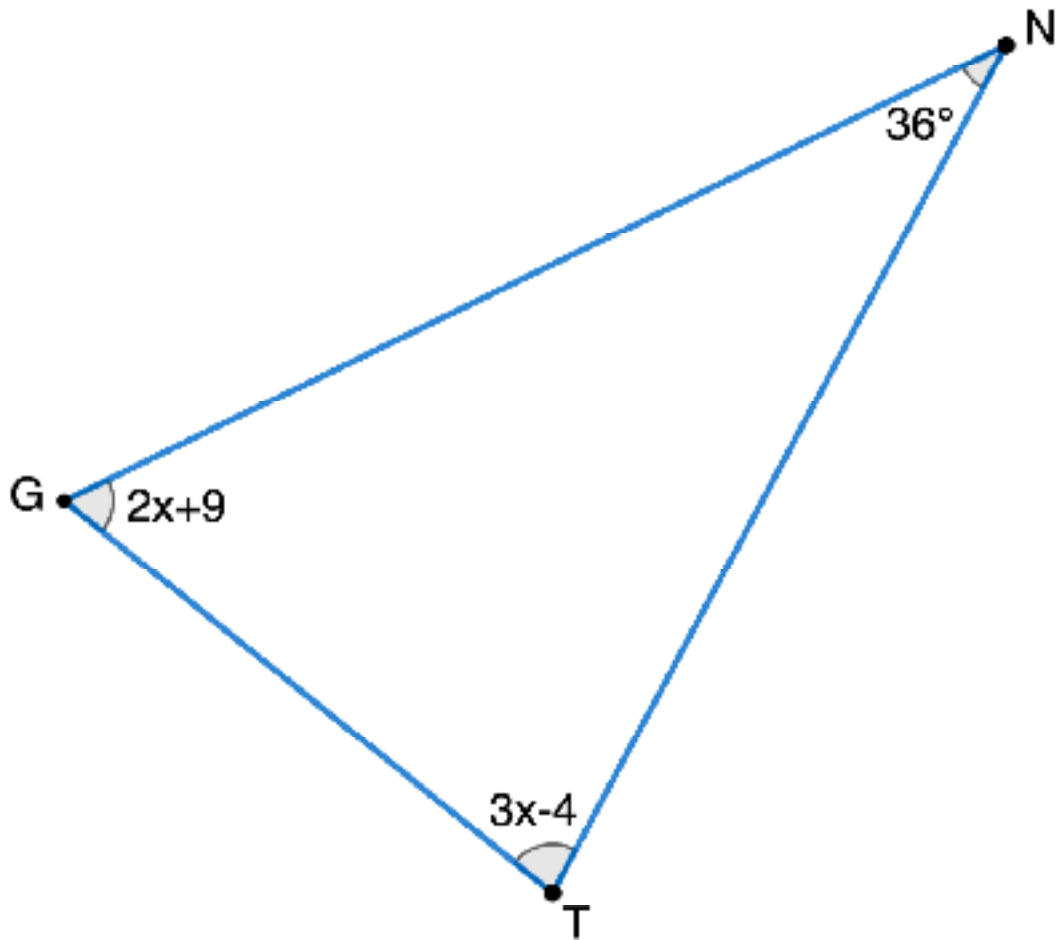
The sum of the measures of the interior angles of a triangle is 180° .

1. Triangle LDK is shown.



- Write an equation that can be used to find the measure of the unknown angle in the triangle.
- Solve the equation to find the measure of $\angle KLD$.

2. Triangle NGT is shown.

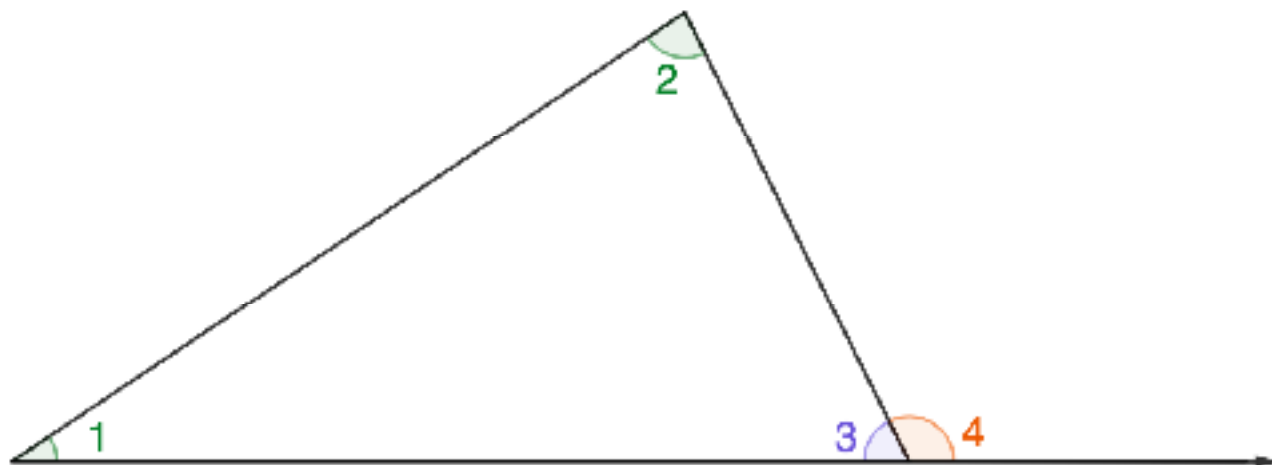


- Write an equation that can be used to find the value of x .
- Solve the equation.
- Find the measure of $\angle GTN$.

13.3.4 Collaboration: Exterior Angles of Triangles

An **exterior angle** is an angle created by extending one leg of a triangle.

In the triangle shown, $\angle 4$ is an **exterior angle** and $\angle 3$ is its **adjacent interior angle**.



Work with your partner to complete the following based on the given information.

1. Complete the statements.

An exterior angle is

- ☐ complementary
- ☐ supplementary
- ☐ vertical

to its adjacent interior angle, so the

sum of $\angle 3$ and $\angle 4$ is

- ☐ 90° .
- ☐ 180° .
- ☐ 360° .

Angles 1, 2, and 3 are the

- ☐ interior
- ☐ exterior

angles of the triangle.

Their sum is

- ☐ 90° .
- ☐ 180° .
- ☐ 360° .

The notation $m\angle 1$ is read as “The measure of angle 1.”

2. Destiny wrote the following mathematical statements on her paper after looking at the triangle above.

$$m\angle 4 + m\angle 3 = 180^\circ$$

$$m\angle 1 + m\angle 2 + m\angle 3 = 180^\circ$$

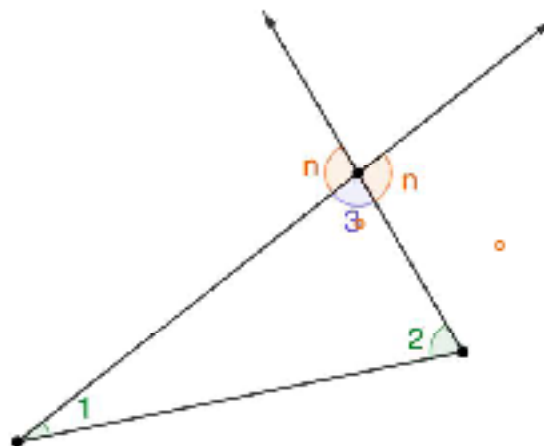
$$m\angle 1 + m\angle 2 = m\angle 4$$

- a. With your partner, discuss Destiny’s reasoning.
- b. Complete the statement.

Any exterior angle of a triangle is equal to

- one of the interior angles of the triangle.
- the sum of the two non-adjacent interior angles of the triangle.

3. Every interior angle of a triangle has two possible exterior angles, depending on the side length that is extended. Consider the triangle shown with both possible exterior angles of $\angle 3$ labeled n° .



- a. Discuss with your partner the angle relationship between both possible exterior angles of $\angle 3$.

b. Work with your partner to complete the statements.

The two possible exterior angles of $\angle 3$ are

- ☐ complementary
- ☐ supplementary
- ☐ vertical

angles, so they both have the same measure.

Therefore, it does not matter which side length is extended to create an

exterior angle because both are equivalent to each other and

- ☐ complementary
- ☐ supplementary
- ☐ vertical

to the adjacent interior angle.

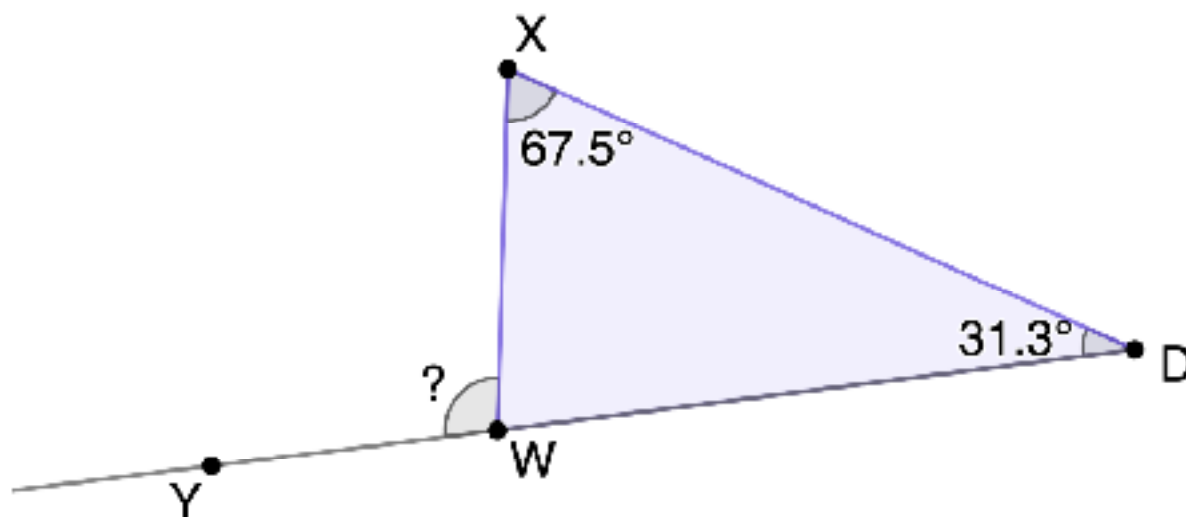
13.3.5 Guided Practice: Exterior Angles of Triangles

The relationship of the exterior angles of a triangle can be used to solve for unknown angle measures in a triangle.



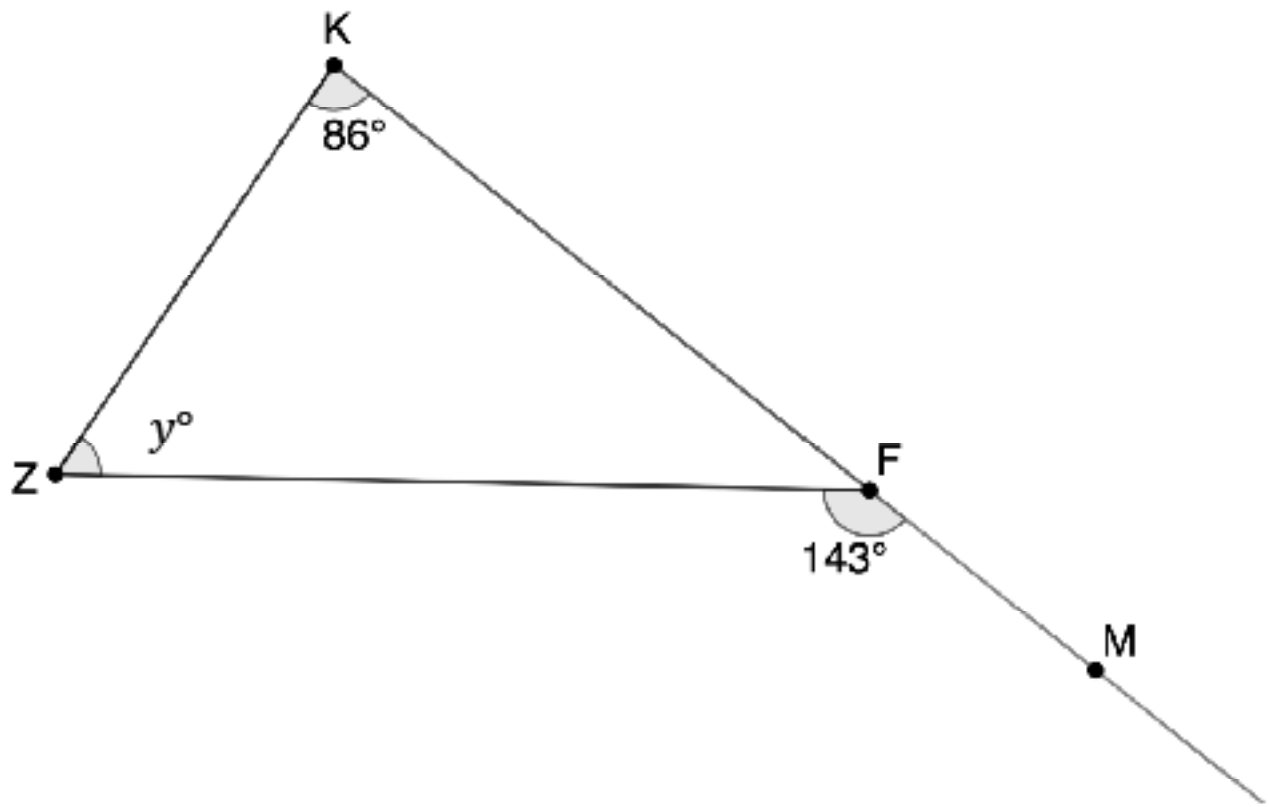
The measure of an exterior angle of a triangle is equal to the sum of the measures of the two non-adjacent interior angles of the triangle.

1. Consider the diagram shown.



Find the measure of $\angle YWX$.

2. A figure is shown.



a. Write an equation that can be used to find the value of y .

b. Solve the equation.

13.3.6 Wrap-Up: Reflection

1. Circle the icon that best describes your understanding of each learning target listed in the table.



I need help.



I got it.



I could help someone.

I can identify the relationship between interior and exterior angles of a triangle.	  
I can write and solve algebraic equations to solve mathematical problems involving angle relationships of triangles.	  